

Haisum Haroon

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OVERVIEW

I study the **fairness**, **reliability** and **security** of intelligent systems, such as **large scale software systems** and **large language models** (llms), through rigorous testing and system development. My recent research focuses on (1) evaluating **LLMs for disability-related bias and fairness** (2) **automated test repair** for software testing. Through my work, I aim to build secure, reliable and human-centered software systems.

SKILLS

Technical: C++, Python, Java, C, SQL, MongoDB, Firebase, Android Development, Web Development (HTML/CSS/JS), OpenAI/LLM APIs.

Research: Experimental Design, User Study Design, Quantitative & Qualitative Data Analysis, Statistical Testing, Thematic Analysis, Dataset Curation, Critical Thinking, Technical Writing.

Selected Coursework: Computer Networks, Operating Systems, Databases, Data Structures, Algorithms, Statistics, OOP, Artificial Intelligence.

RESEARCH EXPERIENCE

Research Assistant — Tufts University *May 2025 – April 2026*

- Conducted the **first systematic study of LLM bias** against autistic communication styles using a **matched-guise paradigm** across three cross-neurotype dialogue scenarios.
- Co-curated a dataset of **300 paired autistic-allistic dialogues** in collaboration with an autistic advisory board, covering indirect speech acts, figurative expressions, and blunt statements.
- Benchmarked **9 state-of-the-art LLMs** across 4 decision-making tasks (hiring, dating, admissions, reasonableness) and 5 prompt conditions, generating **54,000+ experimental data points**.
- Utilized **ordinal logistic regression** with likelihood-ratio tests and **FDR correction** to quantify rating disparities, finding autistic responses rated lower in up to 75% of cases.
- Analyzed **reasoning traces** from DeepSeek-R1 to surface qualitative patterns in how models form judgments under neurodiversity-informed prompting.

Research Assistant — UIUC+ Research Program *Mar 2025 – Present*

- Selected from **1,500+ international applicants** for a research program in software engineering.
- Conducted an in depth **literature review** on **automated test repair** and **software testing**, including papers from FSE, ISSTA, and USENIX.
- Implemented and evaluated automated testing pipelines, analyzing **repair reliability**, **obsolete test detection**, **test coverage**, **mutation behavior**, and **flaky tests**; studied techniques such as **symbolic** and **concolic execution**.
- Wrote reproducibility scripts in Python to validate results across multiple testing frameworks.

Research Assistant — Dr. Faisal Bukhari, AIT/Punjab University *Jan 2025 – April 2025*

- Conducted a **mixed-methods study** on a mental health chatbot; recruited 50 participants via simple random sampling and collected 3–20 day interaction data.
- Gathered structured survey responses on emotional support, satisfaction, usability, demographics.
- Applied multiple statistical tests: **Wilcoxon signed-rank**, **Mann–Whitney U**, **McNemar’s**, **Chi-square**, **Kruskal–Wallis H**, **Spearman’s rank correlation** etc.
- Analyzed relationships between age, education, gender, retention, interaction time, perceived helpfulness, and effectiveness of chatbot against different mental health disorders.

Research Assistant — Dr. Faisal Bukhari, AIT/Punjab University *Dec 2023 – Jul 2024*

- Led a user study of 40 participants on privacy practices with **hands-on inspection of participants’ devices** to assess settings, permissions, and data-sharing behaviors across **six privacy domains**.
- Designed and administered surveys, performed preprocessing and applied **quasi-Poisson regression modeling** to analyze trends.
- Identified **gender-based behavior–perception gaps** and presented findings for privacy education and policy.

PROGRAMMING PROJECTS

XenVault — C++, Python, Flask, HTML/CSS, JavaScript, SQLite

- Developed a **fault-tolerant distributed storage system**
- Implemented **TCP/IP socket programming** with **Winsock2** for robust inter-node communication.

- Engineered a **dynamic chunking** mechanism and metadata management with **XOR encryption** to distribute files across multiple nodes. Developed **reassembly maps** for accurate file reconstruction and ensured data integrity with checksums.
- Built **web interface** supporting file uploads, retrieval, and deletion.

DriveSense — *Android Studio, Java*

- Built a **driving behavior monitoring app** that detects unsafe events including harsh braking, sharp turns, over-speeding, and rapid acceleration.
- Implemented **real-time event detection logic** using accelerometer, GPS, and speed sensors.
- Tracked trips in **persistent internal storage**, recording timestamps, geolocation, and violations.
- Visualized trips and designed a **multi-screen UI** with banner/interstitial ads for monetization using **Google AdMob** and **Google Maps API**

Apparel Management System — *React.js, Tailwind CSS, ASP.NET, C#, SQL, WebRTC*

- Developed a **full-stack web platform** featuring secure authentication, inventory/order management, and role-based admin and customer dashboards.
- Integrated an **AI-powered chatbot** to enhance customer experience and implemented **WebRTC video conferencing** for live online fashion courses.
- Ensured **scalable, secure hosting** with real-time updates and efficient transaction workflows.

TriviumElite — *Firebase, Java, Android Studio*

- Built an **educational quiz app** offering adaptive difficulty and multi-subject support.
- Implemented **secure authentication** and real-time leaderboard using **Firebase**.
- Added performance analytics and **API-based dynamic question fetching**.

EDUCATION

Information Technology University (ITU), Lahore, Pakistan

Bachelor of Science in Computer Science, Expected Graduation: 2027

HONORS AND ACHIEVEMENTS

Selected Research Assistant, UIUC+ Program (5% acceptance rate)

2025